

Experimental Analysis of Public Policies

Session 6

Event Studies and Staggered DiD

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Session Objectives

By the end of this session, students should be able to:

- define event time and interpret omitted periods, leads, and lags;
- read an event-study graph as both diagnostic evidence and effect dynamics;
- explain why traditional TWFE event-study coefficients need qualification;
- identify anticipation, dynamic effects, and weak support in long lags;
- explain why staggered adoption complicates TWFE;
- distinguish never-treated, not-yet-treated, and already-treated comparison units;
- define $Y_{it}(g)$ and interpret $ATT(g, t)$;
- describe why modern DiD estimators keep cohort and event time explicit.

Roadmap

From 2x2 DiD To Event Time

Event Time And Traditional TWFE

Staggered Adoption: Who Is Control?

Why Naive TWFE Can Mislead

Group-Time ATT And Aggregation

Practice And Bridge

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From DiD To Dynamics

- Session 5: one treated group, one comparison group, one treatment date.
- Core assumption: the comparison group reveals the treated group's untreated trend.
- New problem: effects may appear slowly, fade, or differ across cohorts.
- New question: who serves as control at each date?

Central question

What changes when treatment timing and treatment effects are heterogeneous?

Essential Versus Optional

Essential	event time, reference period, leads, lags, support, comparison groups
Optional	full negative-weight algebra and estimator derivations

Teaching choice

Students should leave with the comparison logic even if they do not master every estimator.

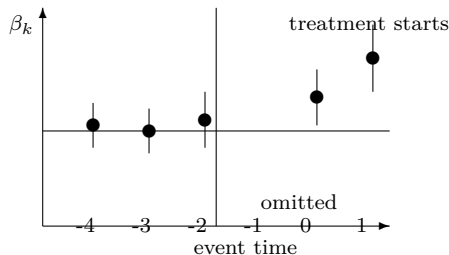
Why Event Studies?

- Visualise pre-treatment dynamics.
- Estimate short-run and medium-run effects.
- Detect possible anticipation.
- Communicate timing in a way tables often cannot.

Interpretation habit

Always name the omitted period before saying what any coefficient means.

Event-Study Graph: What To Read



- Reference period.
- Pre-treatment coefficients.
- Change around treatment onset.
- Confidence intervals and support.

Event-Study Workshop: Read The Graph

Prompt: an event-study graph shows coefficients from $k = -5$ to $k = 5$, omitting $k = -1$.

1. What is the reference period?
2. Do the leads suggest pre-trends, anticipation, noise, or low power?
3. Does the effect appear immediately, gradually, or temporarily?
4. Which event times are likely based on fewer cohorts?

Student output

One sentence on pre-trends, one on effect timing, one on uncertainty or support.

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Event Time

$$k = t - G_i$$

- G_i : first period in which unit i receives treatment.
- $k = 0$: first treated period.
- $k = -1$: period just before treatment.
- $k = 1$: one period after treatment.
- Never-treated units have $G_i = \infty$, so event time is not defined for them.

Event-Time Data Layout

	2022	2023	2024	2025
Early adopter, $G = 2023$	$k = -1$	$k = 0$	$k = 1$	$k = 2$
Late adopter, $G = 2025$	$k = -3$	$k = -2$	$k = -1$	$k = 0$
Never-treated, $G = \infty$	control	control	control	control

Student output

Convert two calendar years into event time and mark which units are untreated in each year.

Traditional TWFE Event-Study Regression

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \in \mathcal{K}, k \neq -1} \beta_k 1[G_i < \infty, t - G_i = k] + u_{it}$$

- Unit fixed effects absorb permanent unit differences.
- Time fixed effects absorb common calendar shocks.
- One event-time period is omitted as the reference.
- Never-treated units have all event-time indicators equal to zero.
- Endpoints such as $k \leq -5$ or $k \geq 5$ are often binned.

Traditional TWFE: Immediate Caveat

Do not overread the dots

In staggered designs with heterogeneous effects, traditional TWFE β_k 's may not equal clean event-time ATTs.

- Already-treated units can enter comparisons for newly treated units.
- Dynamic effects from early cohorts can contaminate lead or lag coefficients.
- The graph still teaches timing, but it needs a comparison-group audit.

The Omitted Period

- Usually omit $k = -1$, the period just before treatment.
- Each coefficient is interpreted relative to that omitted period.
- Changing the omitted period changes the visual reference point, not the underlying information.

Reading cue

If $k = -1$ is omitted, a coefficient at $k = 2$ is relative to one period before treatment.

Leads And Pre-Trend Diagnostics

- Lead coefficients correspond to periods before treatment.
- In a clean comparison design, leads should be close to zero under no anticipation and parallel pre-trends.
- In traditional staggered TWFE, leads may also reflect contamination from heterogeneous post-treatment effects.
- They are diagnostics, not definitive tests.

Low-power warning

A flat pre-period with wide confidence intervals may simply mean the design cannot detect violations.

Lags And Dynamic Effects

- Lag coefficients correspond to periods after treatment.
- In a clean event-study estimator, lags describe post-treatment dynamics.
- In traditional staggered TWFE, lags need a comparison-group and weighting audit before causal interpretation.
- Long-run lags may be imprecise if few cohorts are observed that long.

Policy example

A training programme may show no effect at $k = 0$, gains at $k = 2$, and fade-out by $k = 5$.

Anticipation

- Units may respond before official treatment starts.
- Firms may invest before a tax change.
- Households may move before a boundary reform.
- Researchers may redefine the treatment date or omit anticipation periods.

Interpretation choice

Sometimes the relevant policy is the announcement plus implementation, not only the legal start date.

How To Read The Graph

1. Locate the omitted period.
2. Identify the estimator and comparison group.
3. Check the pre-treatment coefficients without treating them as proof.
4. Look for a change around treatment onset.
5. Compare magnitudes to policy relevance.
6. Notice confidence intervals and sample support.

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Example: Minimum Wage

- Cities raise minimum wages in different years.
- Employment may adjust slowly.
- Early adopters may differ politically and economically.
- Later adopters can be untreated controls before they adopt, but not after.

Class question

Should a city that raised the minimum wage three years ago serve as a control for a newly treated city?

Staggered Adoption

- Different units adopt the policy in different periods.
- Some units may never adopt.
- Some treated units may act as controls for later-treated units before they adopt.
- Already-treated units may also enter traditional TWFE comparisons.

Why it matters

Control status becomes relative to calendar time and treatment cohort, not a fixed label.

Cohort-By-Calendar Grid

Group	2022	2023	2024	2025
Early adopters	untreated	treated	treated	treated
Late adopters	untreated	untreated	untreated	treated
Never-treated	untreated	untreated	untreated	untreated

Student output

For 2024, mark who is treated, who is not-yet-treated, who is never-treated, and who should not be used as an untreated control.

Staggered Adoption: Debrief Grid

Comparison			Interpretation risk
Treated treated	vs	never- treated	clean if never-treated units are valid controls
Treated treated	vs	not-yet- treated	useful before later adoption occurs
Treated treated	vs	already- treated	risky when effects evolve over time
Early vs late adopters			may mix cohort differences and dynam- ics

Reflex

In staggered DiD, always ask: who is serving as the control group at each date?

Comparison Groups

- Never-treated units are often the cleanest comparison group if they are credible.
- Not-yet-treated units can be useful before they adopt.
- Already-treated units are problematic if treatment effects evolve over time.
- The comparison group determines the estimand and the external-validity claim.

Diagnostic prompt

For each event time, ask whether the comparison units are never-treated, not-yet-treated, or already exposed.

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TWFE Weights: Intuition

- TWFE combines many 2x2 comparisons.
- In staggered designs, some comparisons may be undesirable.
- If effects differ by cohort or time since treatment, the weighted average can be hard to interpret.
- The issue is not fixed effects themselves; it is the comparison structure.

Plain words

A single TWFE coefficient can mix early-vs-late, late-vs-early, and treated-vs-never comparisons.

Negative Weights

Advanced warning

With staggered adoption and heterogeneous effects, TWFE can place negative or unintuitive weights on some comparisons.

- This can make the coefficient differ from a transparent average treatment effect.
- The full algebra is optional.
- The essential habit is to mistrust a coefficient whose comparison groups are opaque.

Heterogeneous Effects

- Effects may differ across cohorts.
- Effects may differ by exposure length.
- Early adopters may be systematically different.
- Treatment quality may change as the policy scales.

Example

Early minimum-wage adopters may be high-income cities, while later adopters may face different labour-market conditions.

Common Mistakes

Watch for these

- Treating insignificant leads as proof of validity.
- Forgetting which period is omitted.
- Interpreting traditional TWFE leads and lags without checking comparisons.
- Mixing already-treated units into comparisons without interpretation.
- Reporting one TWFE coefficient when dynamics are central.
- Ignoring support for long-run event times.

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Potential Outcomes With Treatment Timing

- $Y_{it}(g)$: potential outcome at time t if unit i first receives treatment in period g .
- $Y_{it}(\infty)$: potential outcome at time t if unit i is never treated.
- $G_i = g$: unit i 's observed treatment cohort.
- This notation makes timing part of the causal object.

Why new notation?

Static $Y_i(1)$ and $Y_i(0)$ hide when treatment begins and how long exposure has lasted.

$$ATT(g, t) = E[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g], \quad t \geq g$$

- g : cohort first treated in period g .
- t : calendar period.
- The target is local to a cohort and date.
- Identification compares cohort g to credible never-treated or not-yet-treated units.

Cleaner Comparison Logic

- Estimate group-time treatment effects.
- Compare treated cohorts to never-treated or not-yet-treated units.
- Keep already-treated units out of untreated control roles.
- Aggregate effects transparently across cohorts and event times.

Teaching cue

Modern estimators are not magic; they make the comparison group and aggregation visible.

Aggregation

- Dynamic aggregation: average by event time.
- Cohort aggregation: average by treatment cohort.
- Calendar-time aggregation: average effects in a given year.
- Overall aggregation: average across treated observations.
- Different aggregations answer different policy questions.

Reporting rule

Never report an aggregate without saying which event times and cohorts receive weight.

Inference And Graph Design

- Cluster at the unit or treatment-assignment level.
- Report confidence intervals for event-time or group-time estimates.
- Long leads and lags can be noisy because support falls at the extremes.
- Graphs should show event time, treatment onset, omitted period, uncertainty, comparison group, and estimator.

Graph warning

Do not interpret each dot as equally precise or equally well supported.

Code Mini-Lab

- Create event-time indicators.
- Estimate a simple traditional TWFE event-study regression.
- Plot coefficients with confidence intervals.
- Compare that representation to group-time ATT logic.
- Make event time visible in the data before fitting the regression.

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Practice Bridge

- Theory sheet: `exercises/theory/session_06_theory.md`.
- QCM: `assessments/qcm/session_06_qcm.md`.
- Exercise: interpret an event-study graph and identify threats.
- Exam skill: explain what a dynamic DiD coefficient compares.

Student deliverable

One paragraph describing leads, lags, omitted period, support, and the relevant comparison group.

Transition To Session 7

- IV identified local effects for compliers.
- Staggered DiD identifies local effects by cohort and calendar time.
- RDD will identify effects local to units around an institutional cutoff.

Bridge intuition

We move from “same trend without treatment” to “smooth potential outcomes at the cutoff”.

Key Takeaways

- Event studies show timing, diagnostics, and dynamic effects.
- Leads are useful but not definitive tests.
- Traditional TWFE event studies need caution under staggered adoption and heterogeneous effects.
- Cleaner estimands keep cohort and event time explicit.
- Always report the comparison group and the aggregation.

Appendix: Names To Recognise

- Sun and Abraham: interaction-weighted event studies.
- Callaway and Sant'Anna: group-time average treatment effects.
- Borusyak, Jaravel, and Spiess: imputation approach.
- de Chaisemartin and D'Haultfoeuille: robust DiD estimators.

How to use this appendix

Recognise estimator names in papers; focus class time on the comparison logic and interpretation.

References and Further Reading

- Sun and Abraham (2021), “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects”.
- Callaway and Sant’Anna (2021), “Difference-in-Differences with Multiple Time Periods”.
- Goodman-Bacon (2021), “Difference-in-Differences with Variation in Treatment Timing”.
- Roth, Sant’Anna, Bilinski, and Poe (2023), “What’s Trending in Difference-in-Differences?”.
- Borusyak, Jaravel, and Spiess (2024), “Revisiting Event-Study Designs: Robust and Efficient Estimation”.
- de Chaisemartin and D’Haultfoeulle (2020), “Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects”.

Reading reflex

When reading a modern DiD paper, identify the comparison group and aggregation before interpreting the headline coefficient.